

Radar-Based Remote Gait Monitoring with Embedded Data Acquisition and Clinician-Facing Metrics

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PROBLEM STATEMENT

- Parkinson's gait disturbances increase fall risk and reduce mobility.
- Clinic visits miss daily variability; subjective scales lack home data.
- Need privacy-preserving longitudinal monitoring for clinician review.

BACKGROUND & MOTIVATION

- Cameras raise privacy concerns and depend on lighting/occlusion.
- Inertial sensors drift and suffer long-term compliance issues.
- 60 GHz mmWave radar is contactless, privacy-preserving, and robust.

OUR APPROACH

- Infineon BGT60TR13C 60 GHz FMCW radar SoC.
- Two configurations: fixed corridor tripod and shoe-mounted radar.
- Five gait styles: regular, slow, fast, festination, and FOG.

RADAR DEPLOYMENT SETUPS

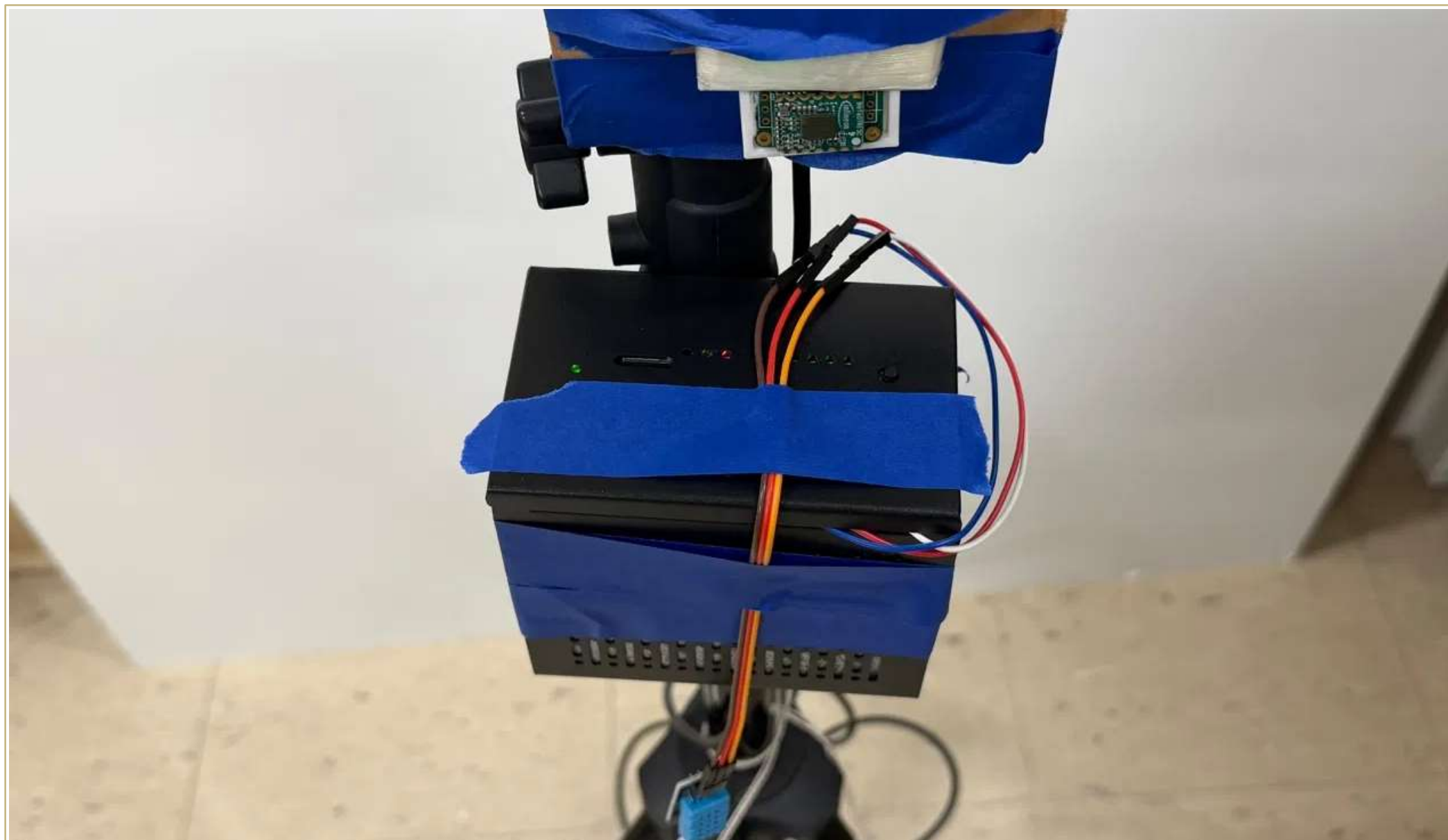


Fig. 1. Tripod-mounted fixed corridor radar - BGT60TR13C observing a 5 m walking corridor.

EMBEDDED ACQUISITION SYSTEM



Fig. 2. Edge acquisition hardware - RPi host supports untethered collection, live step count, and run-folder packaging.

SIGNAL PROCESSING PIPELINE

- FMCW Frame Acquisition**
128 chirps/frame - 12.9 fps - Raspberry Pi 5 USB stream
- Range-Doppler Map**
2D FFT -> range and Doppler profiles per frame
- Clutter Suppression**
Background subtraction + Doppler-domain filtering
- Velocity-Time (VT) Map**
Range gating to ROI - power-weighted velocity profile
- Step Detection & Metrics**
MAD envelope + adaptive peaks -> steps, cadence, CV, MAPE

FIXED CORRIDOR RADAR - RESULTS

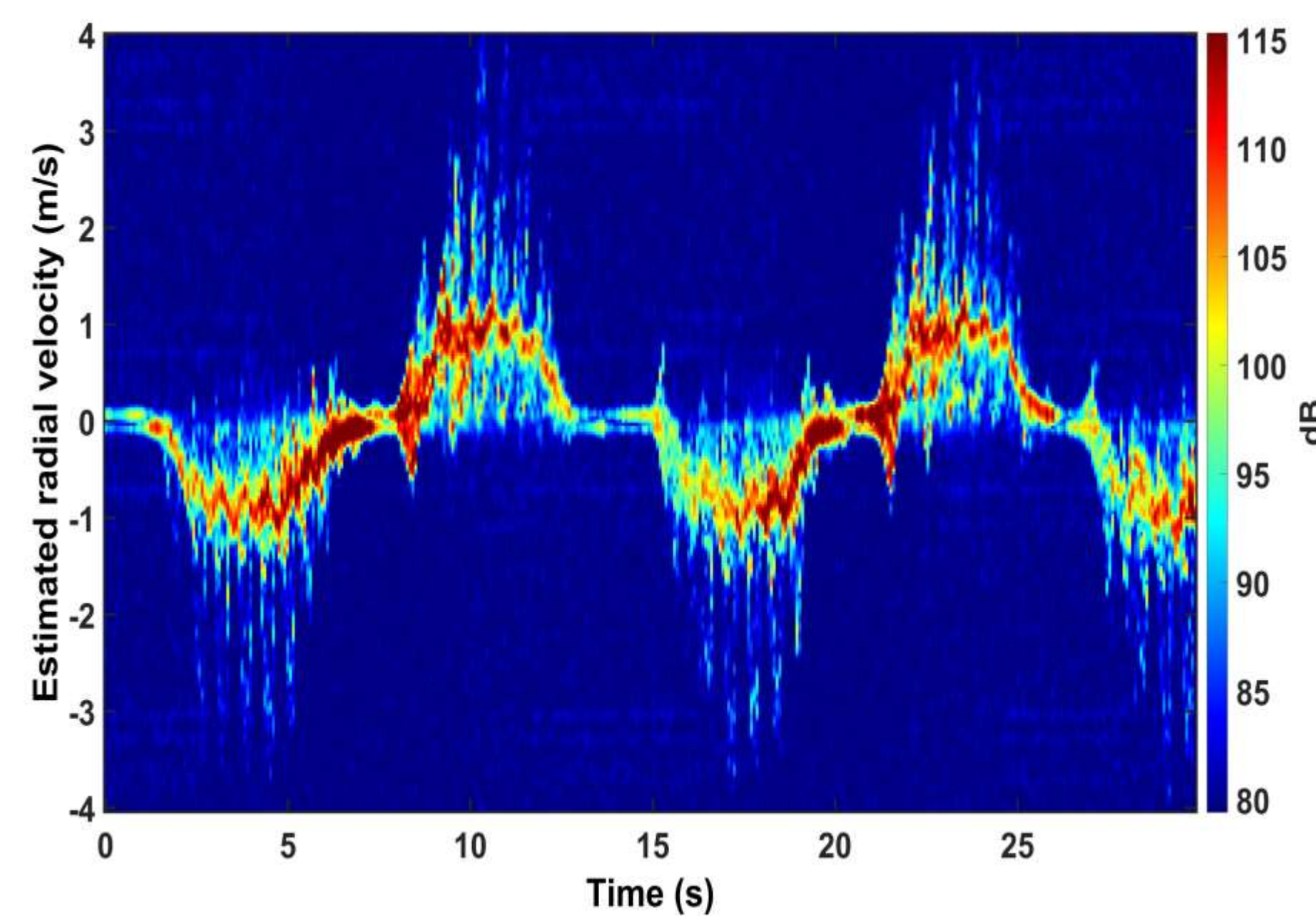


Fig. 3. Fixed corridor VT map (P3). Bulk-motion Doppler dominates; step micro-Doppler appears as weaker modulations.

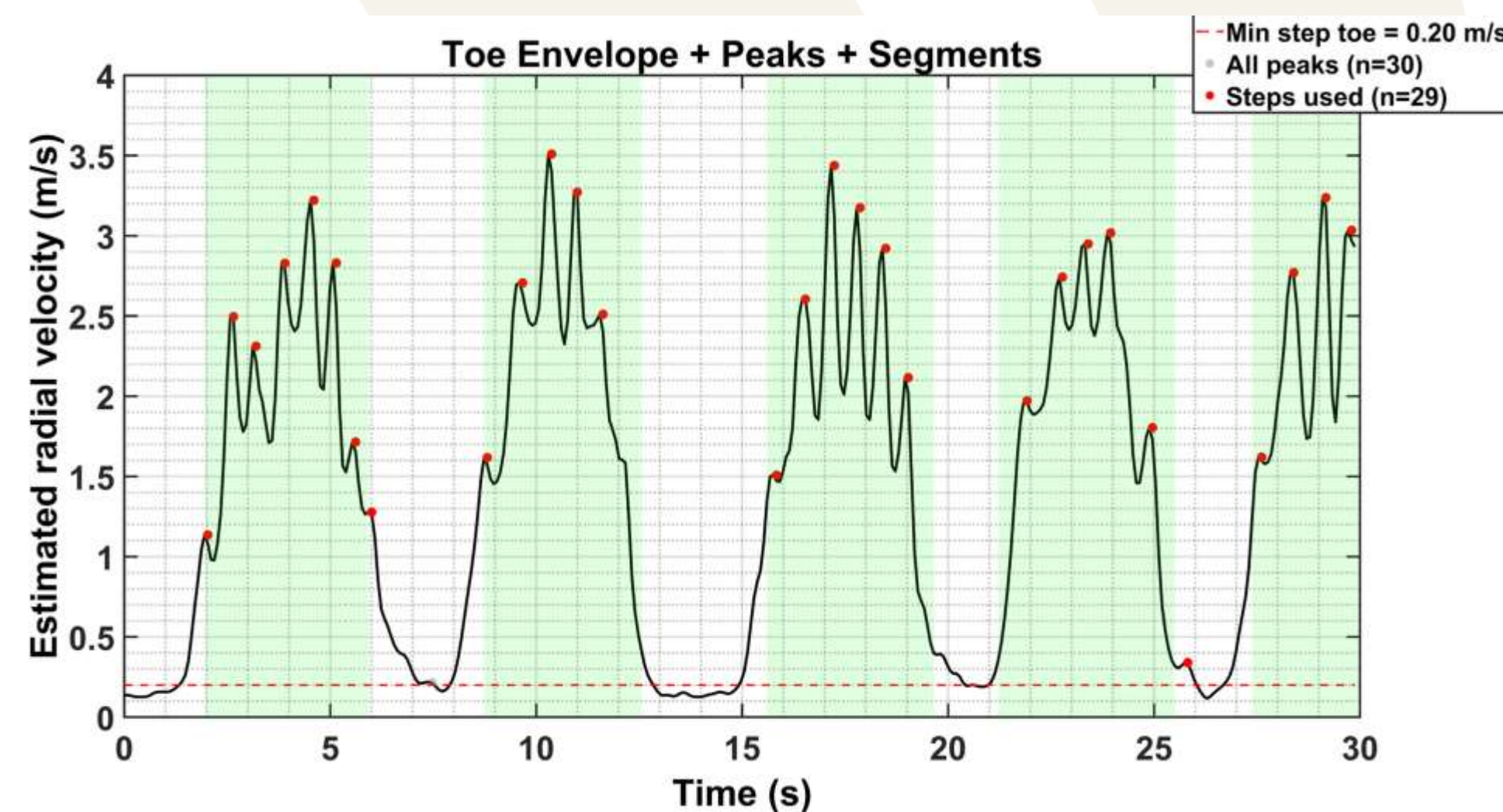


Fig. 4. Fixed corridor foot-velocity envelope. Step events are detected from thresholded velocity peaks; performance summarized at right.

EVALUATION METRIC

Mean Absolute Percentage Error (MAPE)

$$\text{MAPE (\%)} = 100 \times |\text{Nradar} - \text{NGT}| / \text{NGT}$$

- Nradar = radar-detected step count; NGT = video ground-truth count.
- Lower MAPE means closer agreement with clinician-reviewed video labels.
- Evaluation cohort: N=4 participants across five gait styles.



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SHOE-MOUNTED RADAR - RESULTS

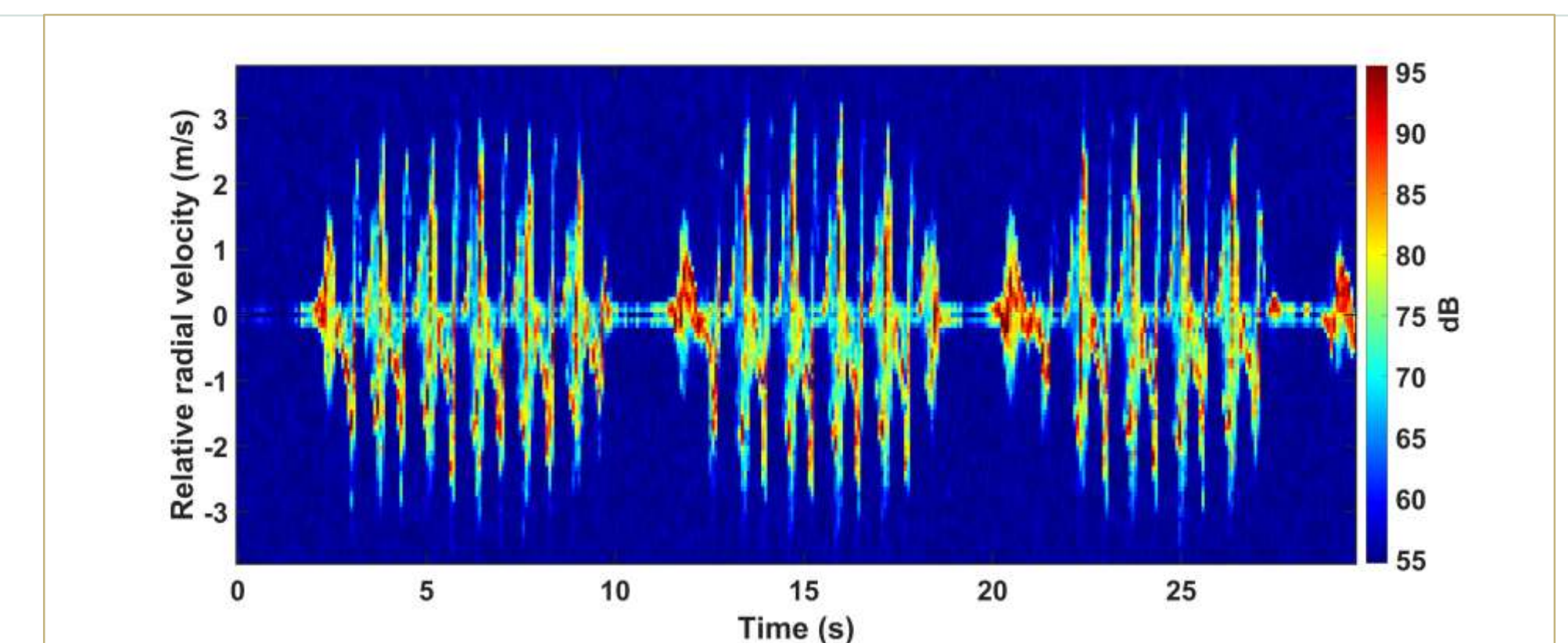
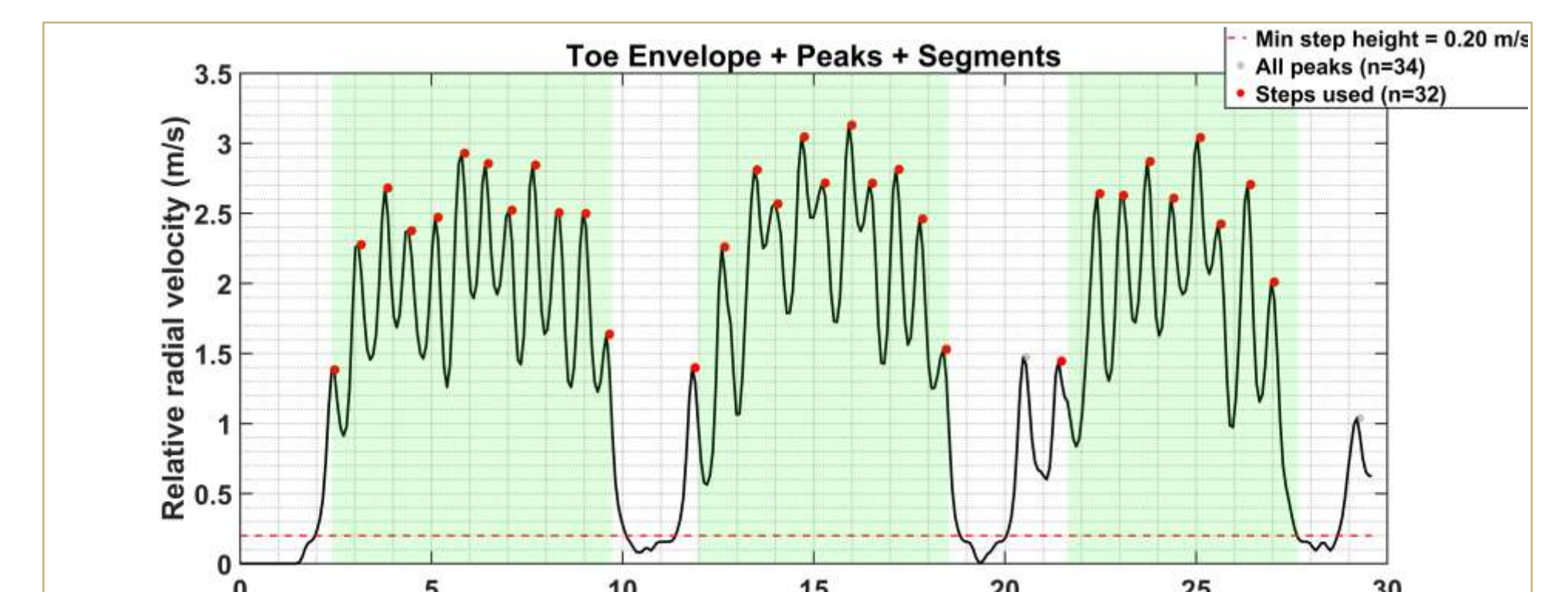


Fig. 5. Shoe-mounted VT map (P1). Clear periodic Doppler bursts from near-field inter-foot motion.



SHOE-MOUNTED SETUP



Prototype sensor on shoe



Future low-profile wearable

Key Insight: Why Shoe-Mounted Outperforms Fixed

Fixed radar captures whole-body translation, so bulk Doppler masks limb micro-Doppler. Shoe-mounted radar emphasizes near-field inter-foot motion, keeping step bursts strong and well separated.

Less clutter

Cleaner bursts

Better metrics

STEP-COUNT ACCURACY

MAPE vs. video ground truth - evaluation N=4, five gait styles

Gait Style	Fixed MAPE	Shoe MAPE	Reduction
Regular	5.39%	2.14%	60%
Slow	13.9%	5.0%	64%
Fast	20.3%	7.8%	62%
Festination	37.1%	9.6%	74% ★
Freezing	14.1%	4.8%	66%

★ largest gain: festination row highlighted

PHYSICIAN-FACING OUTPUTS

Prototype software view for physician review

Step count + cadence
daily mobility load

Step-time CV
gait variability trend

FOG / festination flags
episodic risk markers

Export summary
clinician note / CSV

Goal: turn raw radar into actionable longitudinal gait metrics, not just engineering plots.

CONCLUSIONS

- Shoe-mounted outperforms fixed across all 5 styles - 60-74% MAPE reduction.
- Near-field inter-foot Doppler gives cleaner step bursts and clinically useful metrics.
- RPi workflow + future wearable enclosure support privacy-preserving home monitoring.